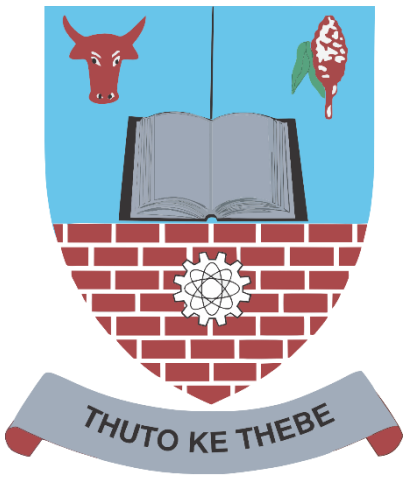


UNIVERSITY of BOTSWANA

CSI435 Assignment 3

Team members:

Thobo Modise 202303550

Lasswell Mahosi 202301477

Tony Moroke 202306319

Tebogo Phiri 202306646

QUESTION 1

a) A supervised language-identification system can distinguish Setswana from another African language, such as isiZulu, by learning patterns from correctly labelled text. The process involves five stages:

- **Labelled examples:** The system is trained using text samples identified as Setswana or isiZulu. For example, “Dumela, o tsogile jang?” is labelled Setswana, while “Sawubona, unjani?” is labelled isiZulu. These examples teach the model the characteristics of each language.
- **Preprocessing:** Text is cleaned by removing HTML tags, duplicate records and irrelevant symbols. This ensures that the model learns from meaningful language patterns rather than unnecessary information.
- **Numerical text features:** Text is converted into numerical representations using techniques such as character n-grams and TF-IDF. These capture spelling patterns and character sequences that help distinguish languages.
- **Classifier:** A supervised algorithm, such as Naive Bayes or Support Vector Machine (SVM), learns the relationship between text features and language labels. It then predicts the language of previously unseen text.
- **Evaluation:** The classifier is tested on held-out examples that were not used during training. Accuracy measures overall correctness, while precision, recall and F1-score help identify errors in distinguishing Setswana from other languages.

Reference: Part 1.

b) Consequences:

Reduced language representation: Mislabelled web pages may cause Setswana text to be incorrectly classified as another language and excluded from training. This reduces the amount of Setswana vocabulary, grammar and sentence structures available to the LLM, weakening its ability to understand and generate Setswana accurately.

Biased model performance:Unequal amounts of training text cause the model to learn well-represented languages more effectively than lower-resource languages. Part 1 reports that Afrikaans and Kiswahili account for approximately half of the collected text, while many other African languages have very limited data. This can result in weaker translation and comprehension performance for languages with fewer examples.

Practical mitigation: Apply automated language identification to verify the labels of collected documents and Use native speakers to review uncertain or incorrectly labelled

samples before including them in the training dataset. This improves the quality of the training data and reduces the likelihood of teaching the model incorrect language patterns.

Reference: Part 1.

QUESTION 2

a) Tokens are smaller units of text, such as words, subwords or character sequences, produced by a tokenizer. Each token is converted into an **embedding**, which is a numerical vector that allows the neural network to process language mathematically and learn relationships between words.

Using fewer tokens per word is important because:

- **Reduced computational cost:** Shorter token sequences require less processing during training.
- **Lower memory requirements:** Fewer tokens reduce the amount of memory needed to process text.
- **Efficient use of training resources:** More African-language words can be processed within the same token budget.

Part 2 reports that Gemma's tokenizer averaged approximately 2.36 tokens per word across the 40 focus languages. Efficient tokenisation is particularly valuable for lower-resource African languages because computing resources and available training data are limited.

Reference: Part 2.

b) A lower training loss means that a model has improved at predicting tokens in its training data. However, this does not guarantee better performance on unfamiliar questions or translation tasks.

Part 2 demonstrates this through a code-mixed model that achieved the lowest training loss but performed poorly on external African-language benchmarks.

For a Setswana question-answering and translation assistant, I would prioritise two evaluations:

- **AfriMMLU accuracy – Question answering:** Measures the proportion of knowledge-based multiple-choice questions answered correctly and helps assess whether the assistant can understand questions and provide correct answers.
- **FLORES ChrF – Translation**: Measures character-level similarity between generated translations and human reference translations and Helps assess the quality of Setswana-English translations.

Limitation of benchmark scores

Benchmark scores may not accurately represent real-world performance. For example, a translation may achieve a high ChrF score while changing the intended meaning of the original sentence. Therefore, benchmark results should be supported by native-speaker evaluations.

Reference: Part 2.

c) Selection of a Model for Initial Deployment

I would select the **plain baseline model** for the first deployment trial because it demonstrated better translation performance and stronger retention of English-language abilities.

Justification

- The plain baseline achieved higher FLORES ChrF scores (29.61 compared with 29.14), indicating better translation performance.
- It also performed better on AfriScience MT (33.54 compared with 30.46).
- However, the religious-cap mix achieved higher AfriMMLU accuracy (0.352 compared with 0.318), suggesting better knowledge retention.

Risk of catastrophic forgetting: Catastrophic forgetting occurs when continued training causes a model to lose previously learned abilities. Part 2 reports that AfriMMLU accuracy declined from 0.381 in the original Gemma model to 0.318 in the plain baseline.

Decrease = 0.381 − 0.318 = 0.063 (6.3 percentage points).

Therefore, although the plain baseline is suitable for an initial translation-focused trial, its question-answering performance should be monitored to prevent further loss of previously learned knowledge.

Reference: Part 2.

QUESTION 3

a) 1. Training time using A100 GPUs

Formula:

$$T = \frac{\text{Total tokens}}{\text{Throughput}}$$

$$T_A = \frac{5{,}000{,}000{,}000}{16{,}700}$$

$$T_A = 299{,}401.20 \text{ seconds}$$

Convert seconds to hours:

$$T_A = \frac{299{,}401.20}{3600}$$
$$\boxed{T_A = 83.17 \text{ hours}}$$

2. Training time using H200 GPUs

$$T_H = \frac{5{,}000{,}000{,}000}{22{,}300}$$
$$T_H = 224{,}215.25 \text{ seconds}$$

Convert to hours:

$$T_H = \frac{224{,}215.25}{3600}$$
$$\boxed{T_H = 62.28 \text{ hours}}$$

3. Throughput speed-up

Formula:

$$\text{Speed-up} = \frac{\text{H200 throughput}}{\text{A100 throughput}}$$
$$= \frac{22{,}300}{16{,}700}$$
$$\boxed{\text{Speed-up} = 1.34\times}$$

$$83.17 - 62.28 = 20.89 \text{ hours}$$
$$\boxed{\text{Time saved} = 20.89 \text{ hours}}$$

b) Faster GPUs may not reduce total completion time proportionally due to operational interruptions. Part 3 reports that the H200 cluster was shared among researchers, resulting in approximately 14 hours of interruptions. Consequently, 62 hours of active training took about 76 hours of elapsed time, reducing the expected speed advantage.

c) Licence verification and attribution: Verify dataset licences, record sources and acknowledge contributors to prevent copyright violations and unauthorised use.

Informed consent: Obtain contributors' permission and explain how their data will be used, stored and shared to protect their rights and culturally sensitive information.

Reference: Part 3.

QUESTION 4

a) The assistant would serve university students and student-services staff by providing information on registration, accommodation, fees and academic services in Setswana or English.

Suitable Setswana-language sources:

1. **University documents:** Approved Setswana translations of student handbooks, registration guidelines and frequently asked questions.
2. **Verified question-answer examples:** Setswana questions and answers covering common student-service enquiries.

Poor coverage: Collaborate with native speakers to translate university materials and develop additional Setswana question-answer examples.

Language mislabelling: Apply automated language identification and native-speaker verification to correct incorrectly labelled text.

The pilot could retrieve information from approved university documents to improve answer reliability (Parts 1 and 4).

b) Automatic measure – Answer accuracy: Measures the percentage of correct responses against verified reference answers using:
Accuracy = (Correct answers ÷ Total questions) × 100.

Human assessment – Native-speaker evaluation: Setswana speakers would rate responses on a five-point scale for grammatical correctness, clarity, cultural appropriateness and preservation of meaning.

Failure mode: Hallucination may cause fluent but incorrect answers, such as false registration deadlines. Responses should therefore be verified against official university information (Part 4).

c) Data permissions and privacy: Use authorised datasets, credit contributors, remove unnecessary personal information and restrict access to confidential student records to protect privacy.**Cultural appropriateness and safety:** Native speakers should review responses for respectful language, while safety testing should identify discriminatory, misleading or harmful advice. Sensitive enquiries should be referred to university staff, following the safety concerns addressed by AfriGuard in Part 4